Fiscal Year:	FY 2014	**Task Last Updated:**	FY 02/05/2014
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Research Opportunities in Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
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City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	02/01/2010	**End Date:**	01/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS (for BASS) NOTE: End date is now 1/31/2016 per NSSC information (Ed., 1/14/2015) NOTE: End date is now 1/31/2015 per NSSC information (Ed., 3/12/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul (Universities Space Research Association) Olson, Sandra (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AD99A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A ground-based experimental and theoretical project for flame spread and extinction over solid materials is proposed with the intention of leading to a flight experiment in the space station. The research will be fundamental in nature but has practical applications to spacecraft fire safety. The objectives are (1) to determine the flame spread and growth characteristics of solids in concurrent flow in reduced gravity, (2) to determine the mechanisms of flame decay and extinction, (3) to establish a high-fidelity numerical model that can simulate both normal and reduced gravity flames, including the interpretation of the results of NASA-STD-6001A test 1. We intend to study both fabric and thick solid samples. This ground-based research is to extend the previous experimental works on solid flame spread in reduced gravity that were almost exclusively limited to very thin samples and to enhance the model capability to a three-dimensional transient formulation. The ground-based experiments will utilize both the normal-gravity and zero-gravity facilities at NASA Glenn Research center. Considerable effort will be spent to plan a space experiment that has to be carried out in the currently available facility in the International Space Station (i.e. CIR). In addition to James T'ien (PI), Dr. Paul Ferkul (NCSER) and Dr. Sandra Olson (NASA GRC) will participate as Co-Is in the project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	During this period, the following tasks have been performed: (1) Prepare the Science Requirement Document (SRD) of GEL for the Science Concept Review (SCR) which is a part of the Solid Fuel Ignition and Extinction (SoFIE) project. The review was held on July 9-10, 2013. The review panel recommended the continuation of the development toward the space experiments. In their report, the panel provided substantial comments and recommendations. To respond to the panels comments, a revised SRD for GEL was prepared. It will facilitate the writing of the consolidated SRD (combining all five project within SoFIE) to be drafted by the SoFIE project scientist Dr. Ferkul. (2) The GEL experiment requires a PMMA spherical sample be instrumented with 5 internal thermocouples. Substantial effort was made to identify a vender who can custom cast the sample with thermocouple rake prepared by us. A scheme of making the rake has been designed and tested. We are currently waiting to the see the prototype be fabricated. (3) Because gas phase radiation is important in combustion in space, a numerical model including flame radiation is needed. Previously we have used the commercial FLUENT code to model for the PMMA sphere combustion without flame radiation. Commercial codes are generally not easy to do detailed data analysis which is crucial in scientific research. To add flame radiation, Makoto Endo has chosen to develop his own combustion code. This work is in progress.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Liao YTT, T'ien JS. "A numerical simulation of transient ignition and ignition limit of a composite solid by a localised radiant source." Combustion Theory and Modelling. 2013;17(6):1096-124. http://dx.doi.org/10.1080/13647830.2013.831486 , Sep-2013